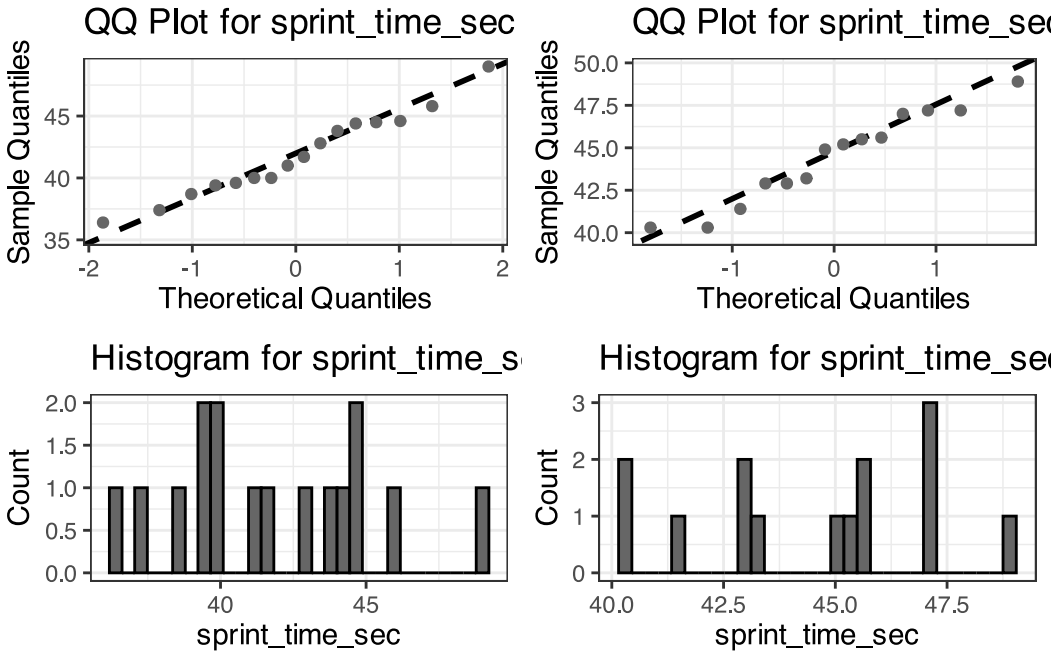
Example 1

- Every cadet runs a timed sprint course as part of camp trials.
- The coaches want us to determine if the Thunder Squadron and Lightning Squadron are finishing the sprint course at different times.
- Checking normality,

```
wonderbolt_camp |> independent_qq(outcome = sprint_time_sec,
                                  grouping = squadron)
```



- Using the `independent_medians()` function,

```
wonderbolt_camp |> independent_medians(outcome = sprint_time_sec,
                                       grouping = squadron,
                                       alternative = "two",
                                       m = 0,
                                       alpha = 0.05)
```

```
Independent medians

M[Lightning Squadron] = 45.05
M[Thunder Squadron] = 41.35
Hodges-Lehmann estimate of median difference: 2.8

95% CI for M[Lightning Squadron] - M[Thunder Squadron]: (0.4, 5.5)
```

```
Hypotheses:
H₀: M[Lightning Squadron] − M[Thunder Squadron] = 0
H₁: M[Lightning Squadron] − M[Thunder Squadron] ≠ 0
Test statistic: W = 168
p-value: 0.019
Conclusion: Reject the null hypothesis (p = 0.019 < α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

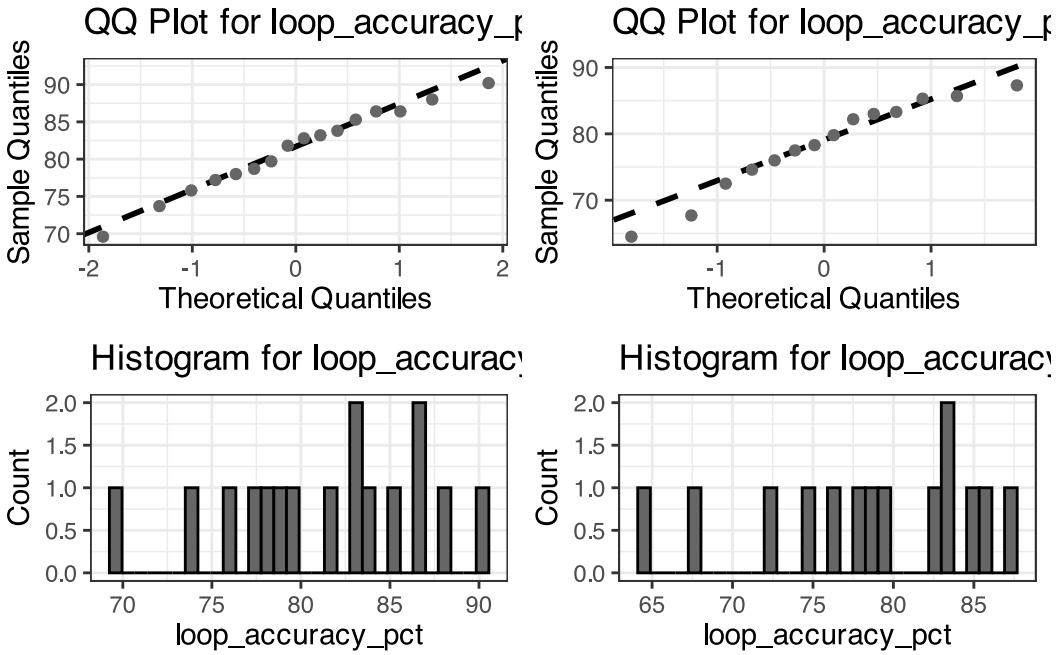
- **Rejection Region**

- **Conclusion and interpretation**

Example 2

- At camp, judges score each cadet's aerial loop maneuver on a 0–100 accuracy scale during trials.
- Based on her gut feeling, the head judge believes that, on average, the Thunder Squadron is performing more accurate loop maneuvers.
- Checking normality,

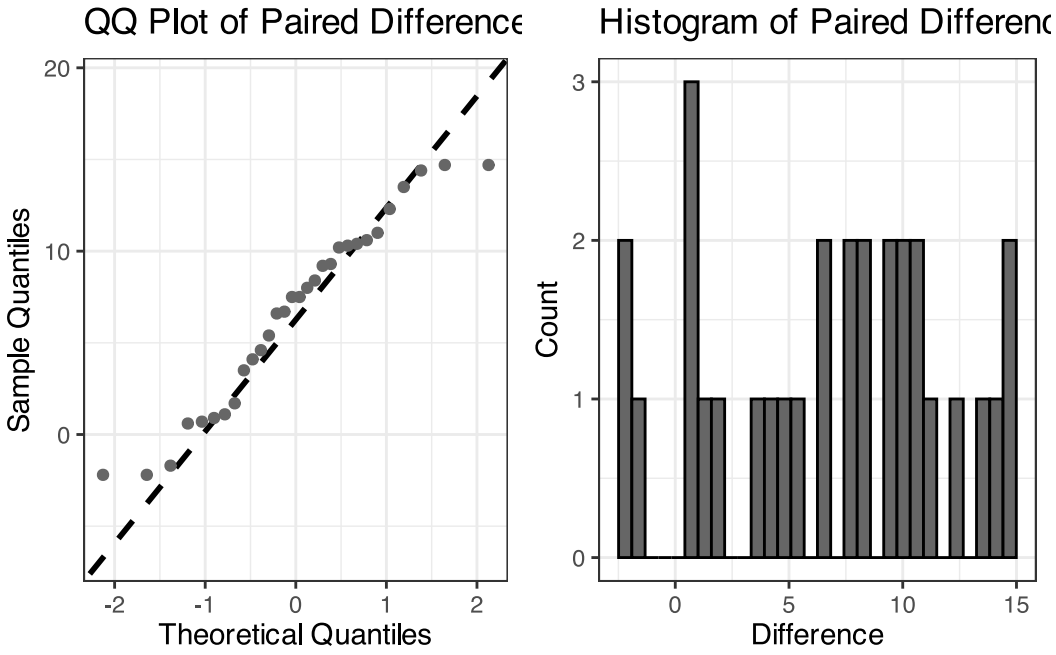
```
wonderbolt_camp |> independent_qq(outcome = loop_accuracy_pct,
                                  grouping = squadron)
```



Example 3

- Each cadet's aerial accuracy is scored when they arrive at camp, and scored again at the end after two weeks of drills.
- Coaches want to know whether cadets are actually flying more accurately by the end of camp.
- Checking normality,

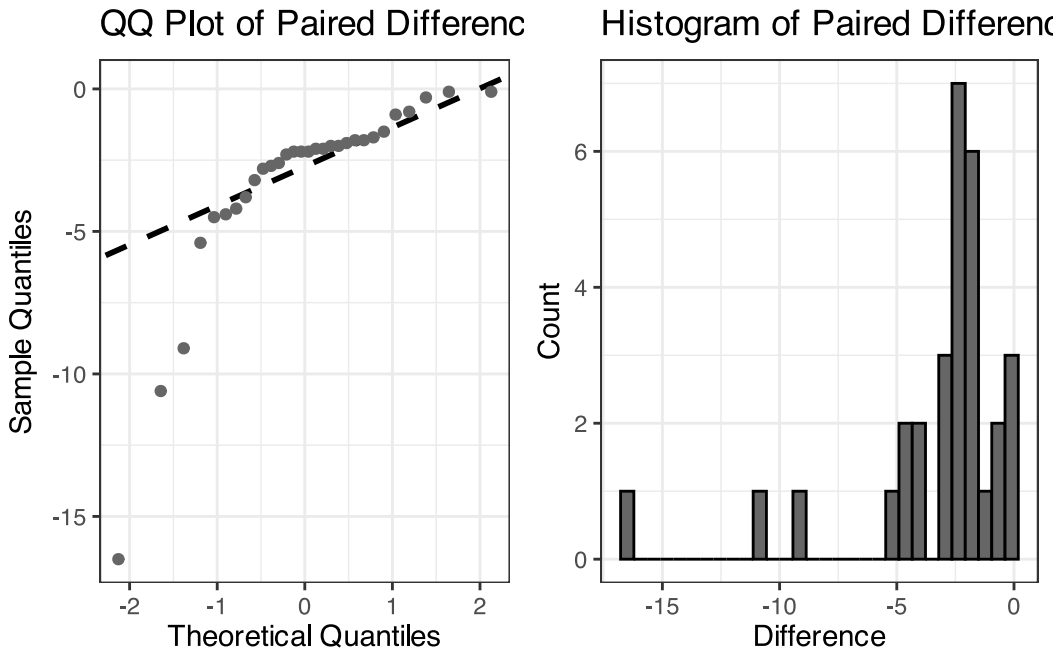
```
wonderbolt_camp |> dependent_qq(col1 = accuracy_score_post,
                                col2 = accuracy_score_pre)
```



Example 4

- Each cadet's recovery time after a hard trial is timed at the start of camp and again at the end.
- Coaches want to know whether cadets are recovering faster by the end of camp.
- Checking normality,

```
wonderbolt_camp |> dependent_qq(col1 = recovery_time_min_post,
                                col2 = recovery_time_min_pre)
```



- Using the `dependent_medians()` function,

```
wonderbolt_camp |> dependent_medians(col1 = recovery_time_min_post,
                                     col2 = recovery_time_min_pre,
                                     alternative = "less",
                                     m = 0,
                                     alpha = 0.05)
```

```
Paired (dependent) medians

Complete pairs: 30
Effective pairs (non-zero differences): 30

M[recovery_time_min_post] = 7.6
M[recovery_time_min_pre] = 12.6
Hodges-Lehmann estimate of median difference: -2.4
```

```
95% CI for M[recovery_time_min_post] − M[recovery_time_min_pre]: (-3.3, -1.95)

Hypotheses:
H₀: M[recovery_time_min_post] − M[recovery_time_min_pre] ≥ 0
H₁: M[recovery_time_min_post] − M[recovery_time_min_pre] < 0
Test statistic: T = 0
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

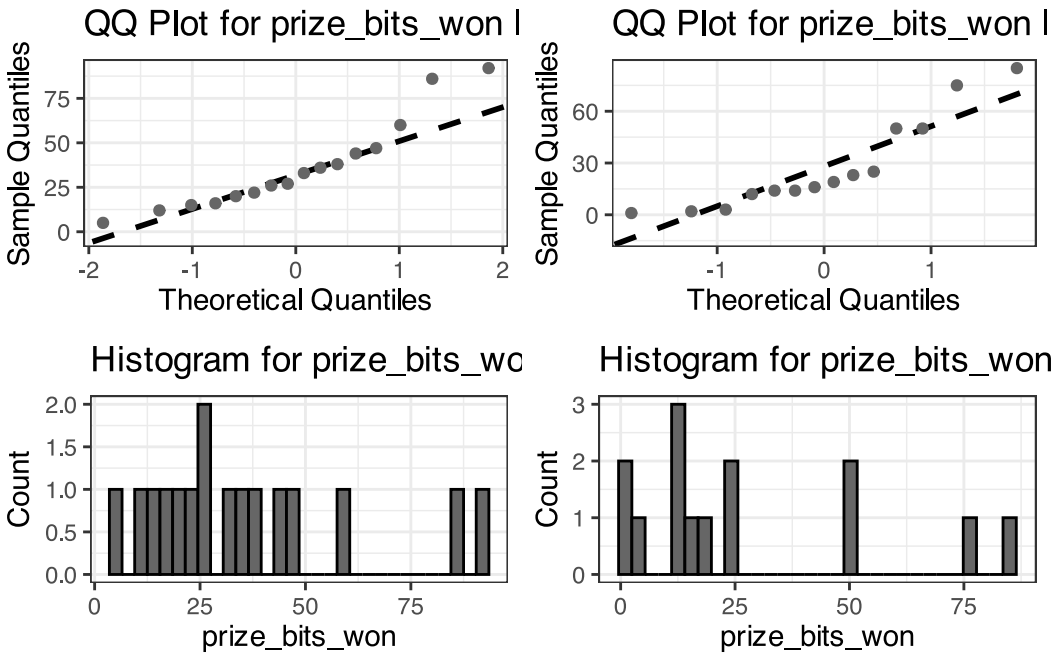
- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 1

- Spectators toss bits to cadets who pull off a standout run during trials.
- Camp organizers want to know if the squadrons are earning prize bits equally.
- Checking normality,

```
wonderbolt_camp |> independent_qq(outcome = prize_bits_won,
                                  grouping = squadron)
```



- Analyzing using the *t*-test,

```
wonderbolt_camp |> independent_means(outcome = prize_bits_won,
                                     grouping = squadron,
                                     alternative = "two",
                                     mu = 0,
                                     alpha = 0.05)
```

```
Independent means (Welch's t-test)

x̄[Lightning Squadron] = 27.7857
x̄[Thunder Squadron] = 36.1875
Point estimate: x̄[Lightning Squadron] - x̄[Thunder Squadron] = -8.4018

95% CI for μ[Lightning Squadron] - μ[Thunder Squadron]: (-27.9469, 11.1433)
```

```
Hypotheses:
H₀: μ[Lightning Squadron] − μ[Thunder Squadron] = 0
H₁: μ[Lightning Squadron] − μ[Thunder Squadron] ≠ 0
Test statistic: t(26.86) = -0.882
p-value: 0.385
Conclusion: Fail to reject the null hypothesis (p = 0.385 ≥ α = 0.050)
```

- Analyzing using the Wilcoxon rank sum,

```
wonderbolt_camp |> independent_medians(outcome = prize_bits_won,
                                        grouping = squadron,
                                        alternative = "two",
                                        m = 0,
                                        alpha = 0.05)
```

```
Independent medians

M[Lightning Squadron] = 17.5
M[Thunder Squadron] = 30
Hodges-Lehmann estimate of median difference: -10

95% CI for M[Lightning Squadron] − M[Thunder Squadron]: (-24, 6)

Hypotheses:
H₀: M[Lightning Squadron] − M[Thunder Squadron] = 0
H₁: M[Lightning Squadron] − M[Thunder Squadron] ≠ 0
Test statistic: W = 79
p-value: 0.176
Conclusion: Fail to reject the null hypothesis (p = 0.176 ≥ α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

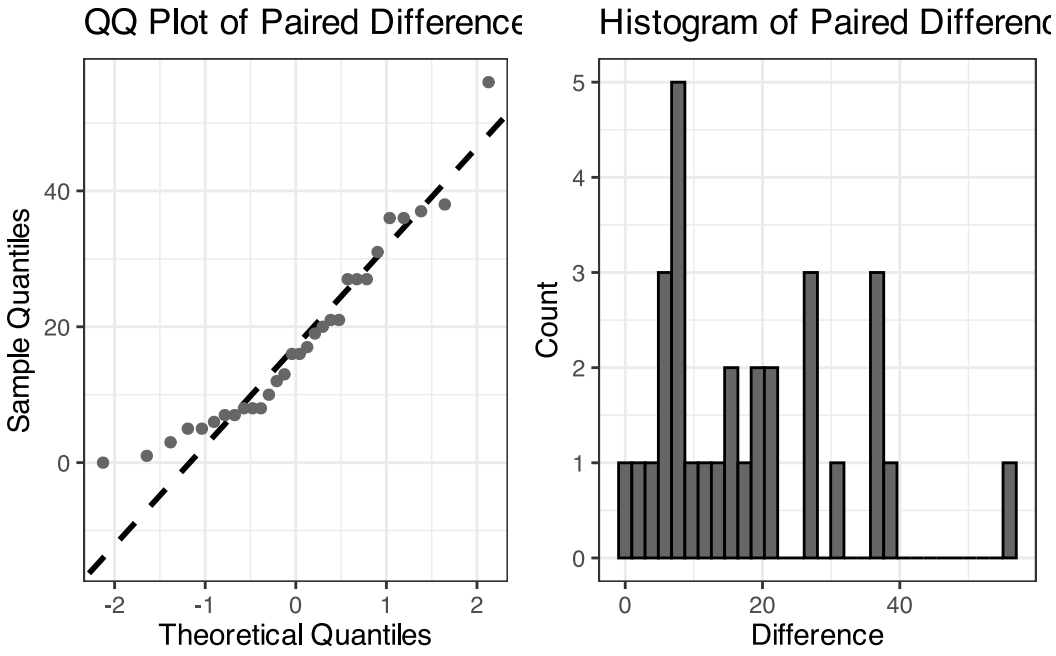
- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 2

- Spectators tip cadets with bits after their runs. The amount tipped is recorded once early in camp and once again near the end.
- Camp organizers want to know whether cadets are earning more in tips as they improve over camp.
- Checking normality,

```
wonderbolt_camp |> dependent_qq(col1 = tip_bits_post,
                                col2 = tip_bits_pre)
```



- Analyzing using the paired *t*-test,

```
wonderbolt_camp |> dependent_means(col1 = tip_bits_post,
                                   col2 = tip_bits_pre,
                                   alternative = "greater",
                                   mu = 0,
                                   alpha = 0.05)
```

```
Paired (dependent) means

Complete pairs: 30
Point estimate: x̄[tip_bits_post] - x̄[tip_bits_pre] = 17.9333
SD of differences: 13.4982
```

```
95% CI for μ[tip_bits_post] − μ[tip_bits_pre]: (12.893, 22.9737)

Hypotheses:
H₀: μ[tip_bits_post] − μ[tip_bits_pre] ≤ 0
H₁: μ[tip_bits_post] − μ[tip_bits_pre] > 0
Test statistic: t(29) = 7.277
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- Analyzing using the Wilcoxon signed rank,

```
wonderbolt_camp |> dependent_medians(col1 = tip_bits_post,
                                      col2 = tip_bits_pre,
                                      alternative = "greater",
                                      m = 0,
                                      alpha = 0.05)
```

```
Paired (dependent) medians

Complete pairs: 30
Effective pairs (non-zero differences): 29

M[tip_bits_post] = 23.5
M[tip_bits_pre] = 9
Hodges-Lehmann estimate of median difference: 17

95% CI for M[tip_bits_post] − M[tip_bits_pre]: (12, 22)

Hypotheses:
H₀: M[tip_bits_post] − M[tip_bits_pre] ≤ 0
H₁: M[tip_bits_post] − M[tip_bits_pre] > 0
Test statistic: T = 464
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

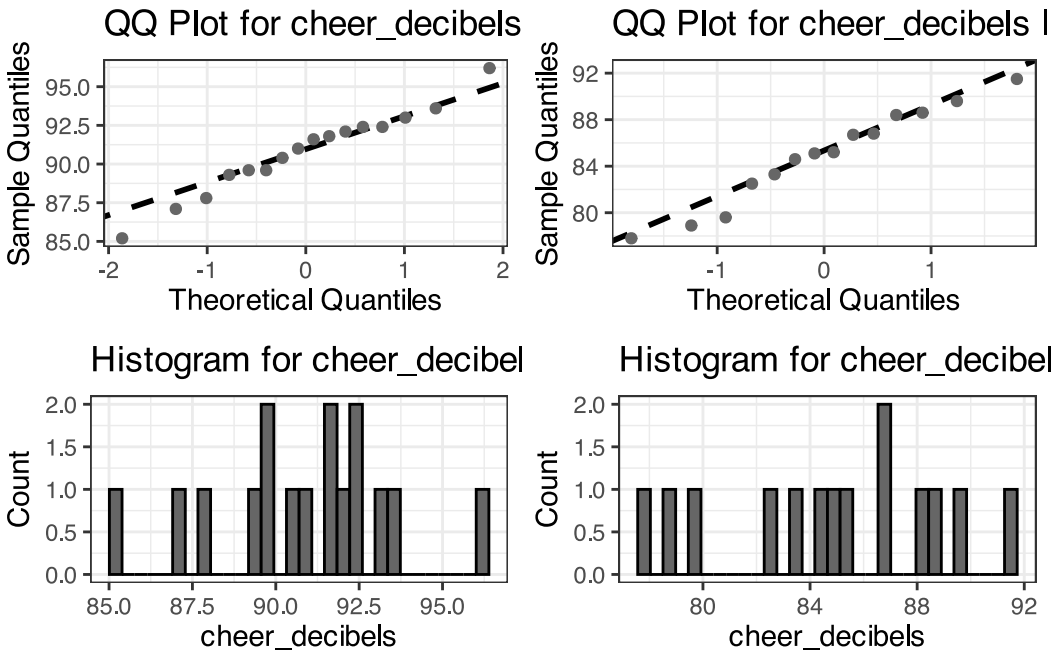
- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 3

- A sound crew records the peak crowd noise (in decibels) during each cadet's run.
- Camp organizers believe that the Thunder Squadron is inspiring a higher level of noise.
- Checking normality,

```
wonderbolt_camp |> independent_qq(outcome = cheer_decibels,
                                  grouping = squadron)
```



- Using the `independent_means()` function,

```
wonderbolt_camp |> independent_means(outcome = cheer_decibels,
                                     grouping = squadron,
                                     alternative = "less",
                                     mu = 0,
                                     alpha = 0.05)
```

```
Independent means (Welch's t-test)

x̄[Lightning Squadron] = 84.9
x̄[Thunder Squadron] = 90.8187
Point estimate: x̄[Lightning Squadron] - x̄[Thunder Squadron] = -5.9187

95% CI for μ[Lightning Squadron] - μ[Thunder Squadron]: (-8.6025, -3.235)
```

```
Hypotheses:
H₀: μ[Lightning Squadron] − μ[Thunder Squadron] ≥ 0
H₁: μ[Lightning Squadron] − μ[Thunder Squadron] < 0
Test statistic: t(21.92) = -4.575
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- Using the `independent_medians()` function,

```
wonderbolt_camp |> independent_medians(outcome = cheer_decibels,
                                       grouping = squadron,
                                       alternative = "less",
                                       m = 0,
                                       alpha = 0.05)
```

```
Independent medians

M[Lightning Squadron] = 85.15
M[Thunder Squadron] = 91.3
Hodges-Lehmann estimate of median difference: -5.65

95% CI for M[Lightning Squadron] − M[Thunder Squadron]: (-8.5, -2.9)

Hypotheses:
H₀: M[Lightning Squadron] − M[Thunder Squadron] ≥ 0
H₁: M[Lightning Squadron] − M[Thunder Squadron] < 0
Test statistic: W = 21.5
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

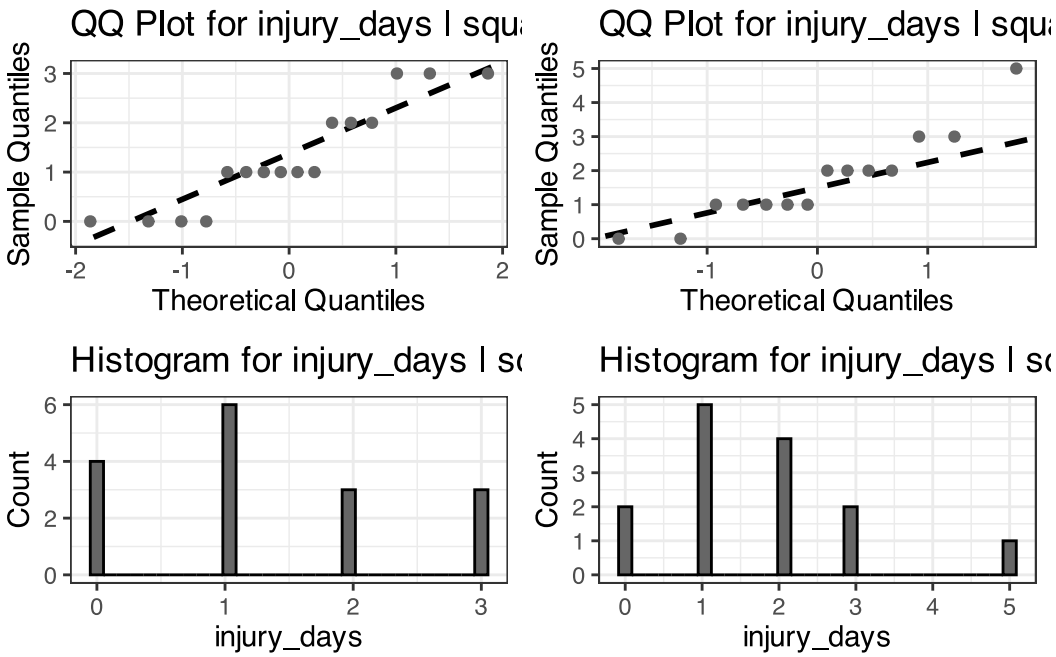
- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 4

- The camp medic tracks how many days each cadet spends recovering from an injury over the two-week camp.
- Coaches want to know if the two training styles lead to the same number of injury days.
- Checking normality,

```
wonderbolt_camp |> independent_qq(outcome = injury_days,
                                  grouping = squadron)
```



- Using the `independent_means()` function,

```
wonderbolt_camp |> independent_means(outcome = injury_days,
                                    grouping = squadron,
                                    alternative = "two",
                                    mu = 0,
                                    alpha = 0.05)
```

```
Independent means (Welch's t-test)

x̄[Lightning Squadron] = 1.7143
x̄[Thunder Squadron] = 1.3125
Point estimate: x̄[Lightning Squadron] - x̄[Thunder Squadron] = 0.4018

95% CI for μ[Lightning Squadron] - μ[Thunder Squadron]: (-0.515, 1.3186)
```

```
Hypotheses:
H₀: μ[Lightning Squadron] − μ[Thunder Squadron] = 0
H₁: μ[Lightning Squadron] − μ[Thunder Squadron] ≠ 0
Test statistic: t(25.11) = 0.902
p-value: 0.375
Conclusion: Fail to reject the null hypothesis (p = 0.375 ≥ α = 0.050)
```

- Using the `independent_medians()` function,

```
wonderbolt_camp |> independent_medians(outcome = injury_days,
                                       grouping = squadron,
                                       alternative = "two",
                                       m = 0,
                                       alpha = 0.05)
```

```
Independent medians

M[Lightning Squadron] = 1.5
M[Thunder Squadron] = 1
Hodges-Lehmann estimate of median difference: 0

95% CI for M[Lightning Squadron] − M[Thunder Squadron]: (-1, 1)

Hypotheses:
H₀: M[Lightning Squadron] − M[Thunder Squadron] = 0
H₁: M[Lightning Squadron] − M[Thunder Squadron] ≠ 0
Test statistic: W = 130
p-value: 0.451
Conclusion: Fail to reject the null hypothesis (p = 0.451 ≥ α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

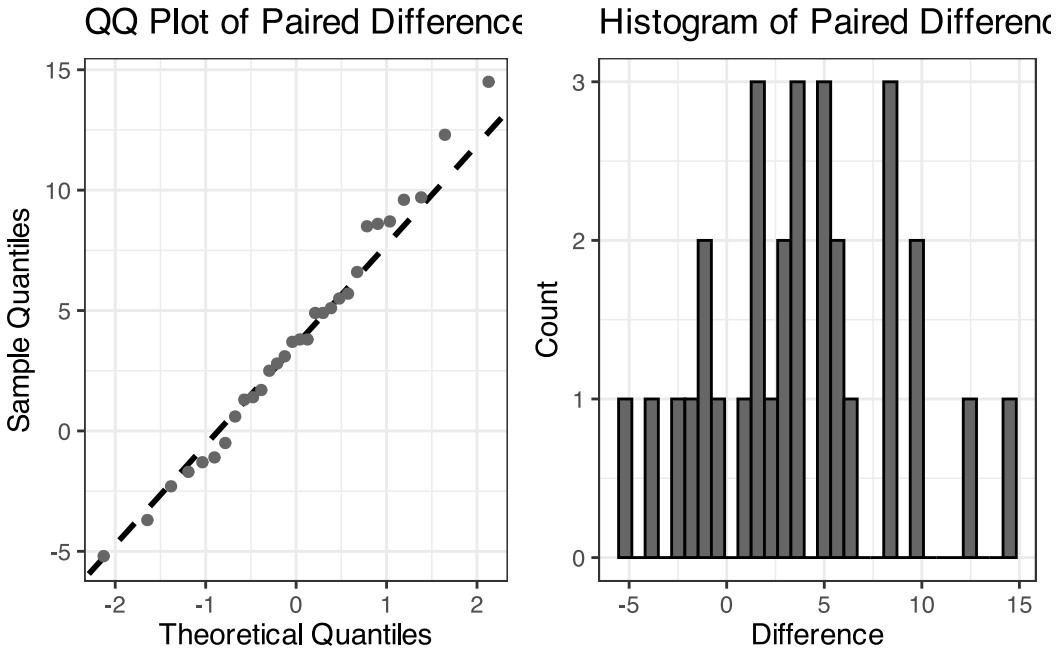
- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 5

- Each cadet's wing-flap rate while hovering in place is measured on arrival, then measured again after two weeks of drills.
- Coaches want to know whether training actually changes a cadet's wing-flap rate.
- Checking normality,

```
wonderbolt_camp |> dependent_qq(col1 = wing_flap_rate_post,
                                col2 = wing_flap_rate_pre)
```



- Using the `dependent_means()` function,

```
wonderbolt_camp |> dependent_means(col1 = wing_flap_rate_post,
                                   col2 = wing_flap_rate_pre,
                                   alternative = "greater",
                                   mu = 0,
                                   alpha = 0.05)
```

```
Paired (dependent) means

Complete pairs: 30
Point estimate: x̄[wing_flap_rate_post] - x̄[wing_flap_rate_pre] = 3.7833
SD of differences: 4.7192

95% CI for μ[wing_flap_rate_post] - μ[wing_flap_rate_pre]: (2.0212, 5.5455)
```

```
Hypotheses:
H₀: μ[wing_flap_rate_post] − μ[wing_flap_rate_pre] ≤ 0
H₁: μ[wing_flap_rate_post] − μ[wing_flap_rate_pre] > 0
Test statistic: t(29) = 4.391
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- Using the `dependent_medians()` function,

```
wonderbolt_camp |> dependent_medians(col1 = wing_flap_rate_post,
                                     col2 = wing_flap_rate_pre,
                                     alternative = "greater",
                                     m = 0,
                                     alpha = 0.05)
```

```
Paired (dependent) medians

Complete pairs: 30
Effective pairs (non-zero differences): 30

M[wing_flap_rate_post] = 55.35
M[wing_flap_rate_pre] = 50.65
Hodges-Lehmann estimate of median difference: 3.7

95% CI for M[wing_flap_rate_post] − M[wing_flap_rate_pre]: (1.9, 5.5)

Hypotheses:
H₀: M[wing_flap_rate_post] − M[wing_flap_rate_pre] ≤ 0
H₁: M[wing_flap_rate_post] − M[wing_flap_rate_pre] > 0
Test statistic: T = 405.5
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and p-Value**

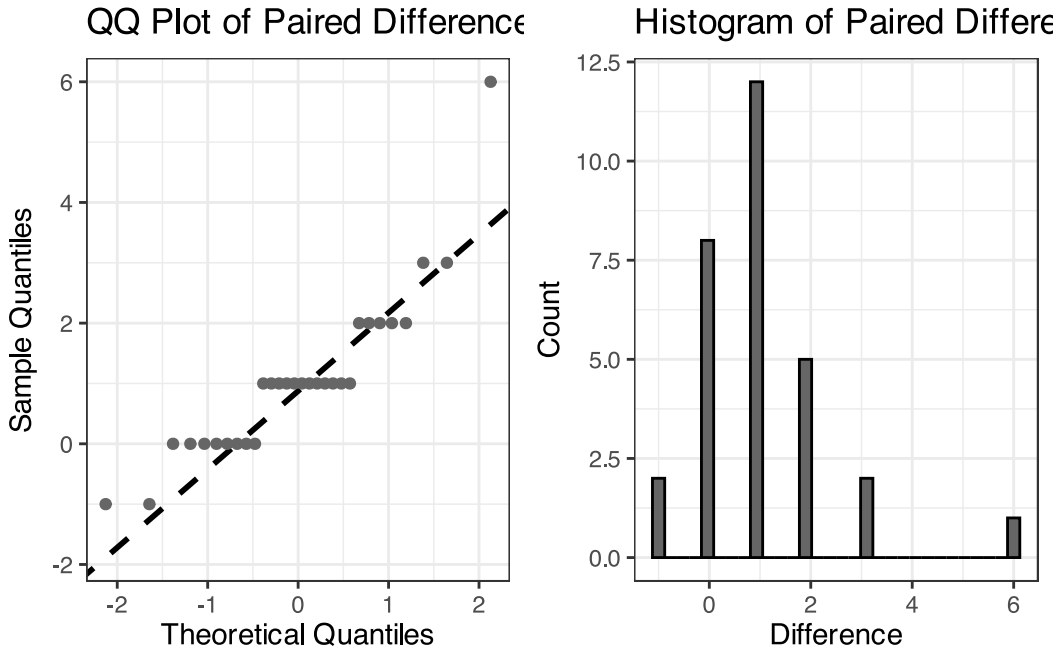
- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 6

- Each cadet rates how confident they feel about their flying on a self-reported scale, once at the start of camp and once at the end.
- Camp staff want to know whether two weeks of training changes how confident cadets feel.
- Checking normality,

```
wonderbolt_camp |> dependent_qq(col1 = confidence_rating_post,
                                col2 = confidence_rating_pre)
```



- Using the `dependent_means()` function,

```
wonderbolt_camp |> dependent_means(col1 = confidence_rating_post,
                                   col2 = confidence_rating_pre,
                                   alternative = "greater",
                                   mu = 0,
                                   alpha = 0.05)
```

```
Paired (dependent) means

Complete pairs: 30
Point estimate: x̄[confidence_rating_post] - x̄[confidence_rating_pre] = 1.0667
SD of differences: 1.3629

95% CI for μ[confidence_rating_post] - μ[confidence_rating_pre]: (0.5578,
```

```
1.5756)

Hypotheses:
H₀: μ[confidence_rating_post] − μ[confidence_rating_pre] ≤ 0
H₁: μ[confidence_rating_post] − μ[confidence_rating_pre] > 0
Test statistic: t(29) = 4.287
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- Using the `dependent_medians()` function,

```
wonderbolt_camp |> dependent_medians(col1 = confidence_rating_post,
                                     col2 = confidence_rating_pre,
                                     alternative = "greater",
                                     m = 0,
                                     alpha = 0.05)
```

```
Paired (dependent) medians

Complete pairs: 30
Effective pairs (non-zero differences): 22

M[confidence_rating_post] = 7
M[confidence_rating_pre] = 6
Hodges-Lehmann estimate of median difference: 1

95% CI for M[confidence_rating_post] − M[confidence_rating_pre]: (0.5, 1.5)

Hypotheses:
H₀: M[confidence_rating_post] − M[confidence_rating_pre] ≤ 0
H₁: M[confidence_rating_post] − M[confidence_rating_pre] > 0
Test statistic: T = 398
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and p-Value**

- **Rejection Region**

- **Conclusion and interpretation**